LESSON 01 / 10

A vector is a displacement, not a location

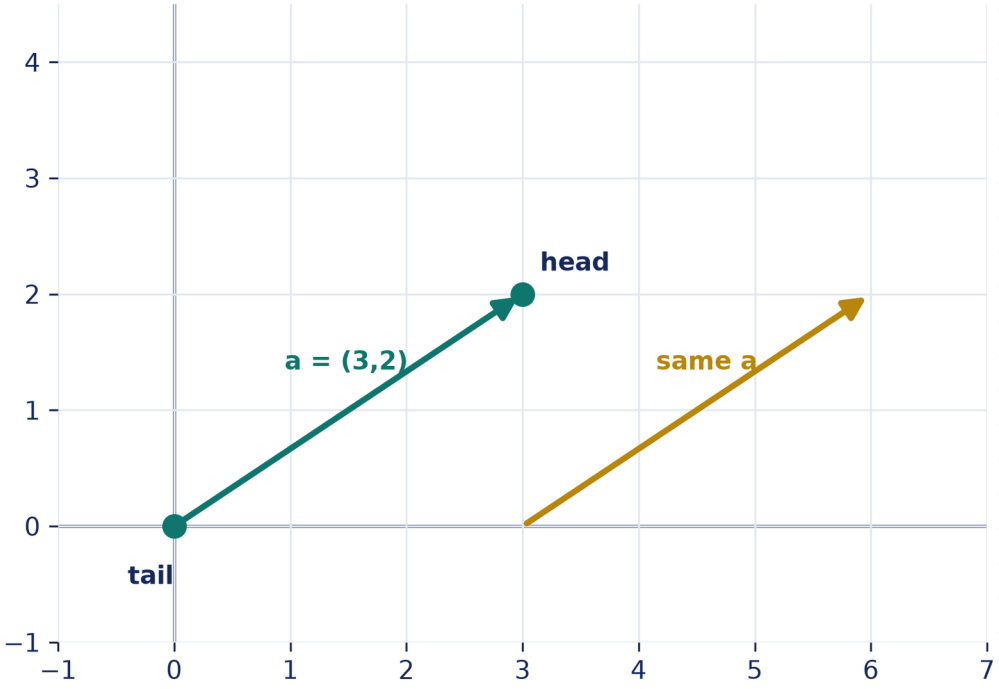
A **scalar** has size only. A **vector** has size and direction. Two arrows represent the same free vector when they have the same length and direction, even if their starting points differ. This freedom to translate an arrow is why vectors simplify geometry.

A **position vector** is different in use: after an origin O is chosen, $\mathbf{a} = \overrightarrow{OA}$ names the point A. A displacement between two points is always

$$\overrightarrow{AB} = \mathbf{b} - \mathbf{a}.$$

Subtract in the order **head minus tail**. Reversing the order reverses the vector.

A free vector can be translated without changing it



READ THE FIGURE
Both arrows have horizontal component 3 and vertical component 2. Their starting points do not change the vector.

Components turn geometry into arithmetic

In three dimensions,

$$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k} = (a_1, a_2, a_3), \qquad |\mathbf{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}.$$

For a nonzero vector, its unit vector is $\hat{\mathbf{a}} = \mathbf{a}/|\mathbf{a}|$. The components of this unit vector are its **direction cosines**, so their squares sum to 1.

The ordinary components are direction ratios: they fix direction but need not have length 1. Normalize before calling them direction cosines.

WORKED EXAMPLE

Magnitude, unit vector and direction cosines

For $\mathbf{a} = (2, -1, 2)$,

$$|\mathbf{a}| = \sqrt{4 + 1 + 4} = 3, \qquad \hat{\mathbf{a}} = \left(\tfrac{2}{3}, -\tfrac{1}{3}, \tfrac{2}{3}\right).$$

Thus the direction cosines are $2/3, -1/3, 2/3$, and

$$\left(\tfrac{2}{3}\right)^2 + \left(-\tfrac{1}{3}\right)^2 + \left(\tfrac{2}{3}\right)^2 = 1.$$

The direction ratios $2, -1, 2$ are not themselves direction cosines.

WORKED EXAMPLE

A displacement is head minus tail

If $A = (1, -2, 3)$ and $B = (4, 2, -1)$, then

$$\vec{AB} = (4 - 1, 2 - (-2), -1 - 3) = (3, 4, -4), \qquad AB = \sqrt{9 + 16 + 16} = \sqrt{41}.$$

Also $\vec{BA} = (-3, -4, 4) = -\vec{AB}$.

Equality, parallelism and the zero vector

Vectors are equal exactly when all corresponding components agree. Two nonzero vectors are parallel when one is a scalar multiple of the other; the scalar's sign tells whether they have the same or opposite direction. Equivalently, their component ratios agree wherever the ratios are defined, but scalar multiplication avoids division-by-zero problems.

The zero vector $\mathbf{0} = (0, 0, 0)$ has length 0 and no direction. Do not divide by $|\mathbf{a}|$ until $\mathbf{a} \neq \mathbf{0}$ is known, and do not assign a unit vector to $\mathbf{0}$.

KEY RESULT

Choose the object before choosing the formula

Use a **position vector** to locate a point, a **displacement** to compare two points, and a **free vector** to represent a direction or physical quantity. Once the object is clear, component arithmetic is routine.

LESSON 02 / 10

Addition follows arrows; scalar multiplication scales them

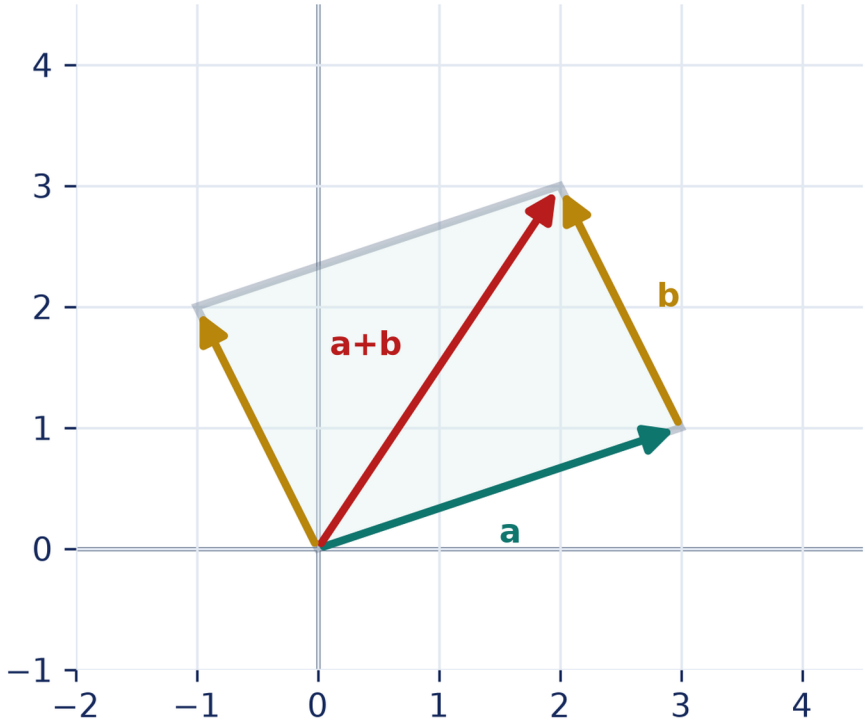
Vectors add componentwise:

$$\mathbf{a}+\mathbf{b}=(a_1+b_1,a_2+b_2,a_3+b_3).$$

Geometrically, place the tail of $\mathbf{b}$ at the head of $\mathbf{a}$. The same resultant is the diagonal of the parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$. Subtraction adds the negative: $\mathbf{a}-\mathbf{b}=\mathbf{a}+(-\mathbf{b})$.

For $\lambda\in\mathbb{R}$, $\lambda\mathbf{a}$ scales length by $|\lambda|$. If $\lambda<0$, it also reverses direction.

The resultant is the parallelogram diagonal



READ THE FIGURE
Tip-to-tail addition and the parallelogram diagonal give the same vector a+b. The shaded region will later be measured by a cross product.

Linear combinations reveal dependence

An expression $\lambda\mathbf{a}+\mu\mathbf{b}+\nu\mathbf{c}$ is a ***linear combination***. Three vectors are linearly dependent if one is a combination of the other two; for three vectors in one plane, this is the algebraic meaning of coplanarity through the origin.

Two nonzero vectors are parallel precisely when $\mathbf{b}=\lambda\mathbf{a}$. In component problems, try scalar multiplication before dividing corresponding components: zero components make ratio shortcuts fragile.

WORKED EXAMPLE

Compute first, then interpret

Let $\mathbf{a} = (3, 1)$ and $\mathbf{b} = (-1, 2)$. Then

$$\mathbf{a} + \mathbf{b} = (2, 3), \qquad \mathbf{a} - \mathbf{b} = (4, -1), \qquad 2\mathbf{a} - 3\mathbf{b} = (9, -4).$$

The parallelogram law is checked immediately:

$$|\mathbf{a} + \mathbf{b}|^2 + |\mathbf{a} - \mathbf{b}|^2 = 13 + 17 = 30 = 2(10 + 5) = 2(|\mathbf{a}|^2 + |\mathbf{b}|^2).$$

Three structural identities save repeated work

For all vectors and scalars,

$$|\mathbf{a} + \mathbf{b}|^2 = |\mathbf{a}|^2 + 2\mathbf{a} \cdot \mathbf{b} + |\mathbf{b}|^2,$$

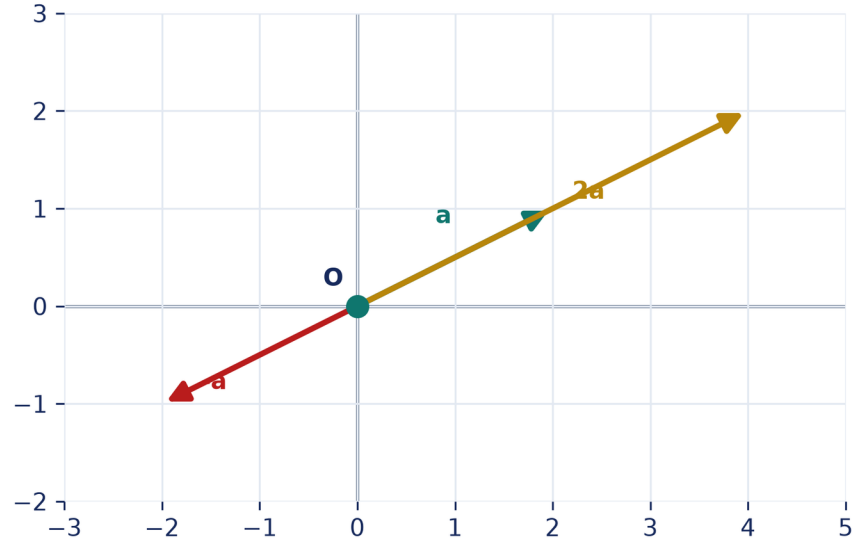
$$|\mathbf{a} + \mathbf{b}|^2 + |\mathbf{a} - \mathbf{b}|^2 = 2|\mathbf{a}|^2 + 2|\mathbf{b}|^2.$$

The second is the ***parallelogram law***: adding the plus/minus expansions cancels the two dot-product terms, so the squares of the diagonals equal the sum of the squares of the four sides. The triangle inequality

$$||\mathbf{a}| - |\mathbf{b}|| \le |\mathbf{a} \pm \mathbf{b}| \le |\mathbf{a}| + |\mathbf{b}|$$

records that one side of a triangle cannot exceed the sum of the other two. Equality cases require parallel directions: for the upper bound, $|\mathbf{a} + \mathbf{b}|$ needs the same direction, while $|\mathbf{a} - \mathbf{b}|$ needs opposite directions.

Scalar multiplication changes length and possibly direction



READ THE FIGURE

The vector 2a has twice the length in the same direction. The vector -a has the same length in the opposite direction.

LESSON 03 / 10

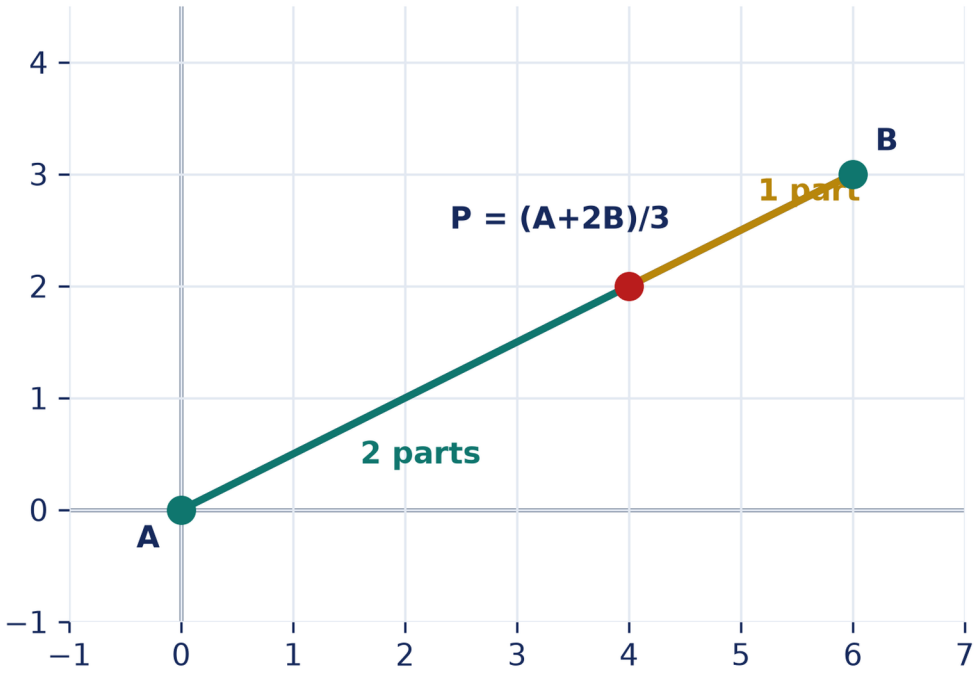
Section formula is a weighted average

Let P divide AB internally in the ratio $AP:PB=m:n$. Because P is $m/(m+n)$ of the way from A to B,

$$\mathbf{p}=\mathbf{a}+\frac{m}{m+n}(\mathbf{b}-\mathbf{a})=\frac{n\mathbf{a}+m\mathbf{b}}{m+n}.$$

The weights are **opposite** the segments: the weight on A is n, and the weight on B is m. The midpoint is the $m=n$ case.

AP:PB = 2:1 means P is two-thirds of the way



READ THE FIGURE
For AP:PB=2:1, P=(A+2B)/3. The coefficient of B is the AP numerator, not the PB numerator.

WORKED EXAMPLE

An internal division in 3D

Divide the join of $A=(1,0,0)$ and $B=(4,5,6)$ in the ratio $2:3$. Then

$$\mathbf{p}=\frac{3(1,0,0)+2(4,5,6)}{5}=\left(\frac{11}{5},2,\frac{12}{5}\right).$$

As a check, $\mathbf{p}-\mathbf{a}=(6/5,2,12/5)$ and $\mathbf{b}-\mathbf{p}=(9/5,3,18/5)$, whose component ratio is $2:3$.

External division needs a signed denominator

If P lies beyond B and $AP:PB=m:n$ with $m>n$, then

$$\mathbf{p}=\frac{m\mathbf{b}-n\mathbf{a}}{m-n}.$$

For $m < n$, the external point lies beyond A. Rather than memorizing two pictures, use directed ratios or derive from $\mathbf{p} = \mathbf{a} + t(\mathbf{b} - \mathbf{a})$ and locate t.

WORKED EXAMPLE

Check an external point with displacements

For $A = (1, -1, 2)$ and $B = (3, 3, -2)$, external division in the ratio $2:1$ beyond B gives

$$P = 2B - A = (5, 7, -6).$$

Indeed $\vec{AP} = (4, 8, -8)$ and $\vec{PB} = (-2, -4, 4)$ have lengths in the ratio $2:1$, with B between A and P.

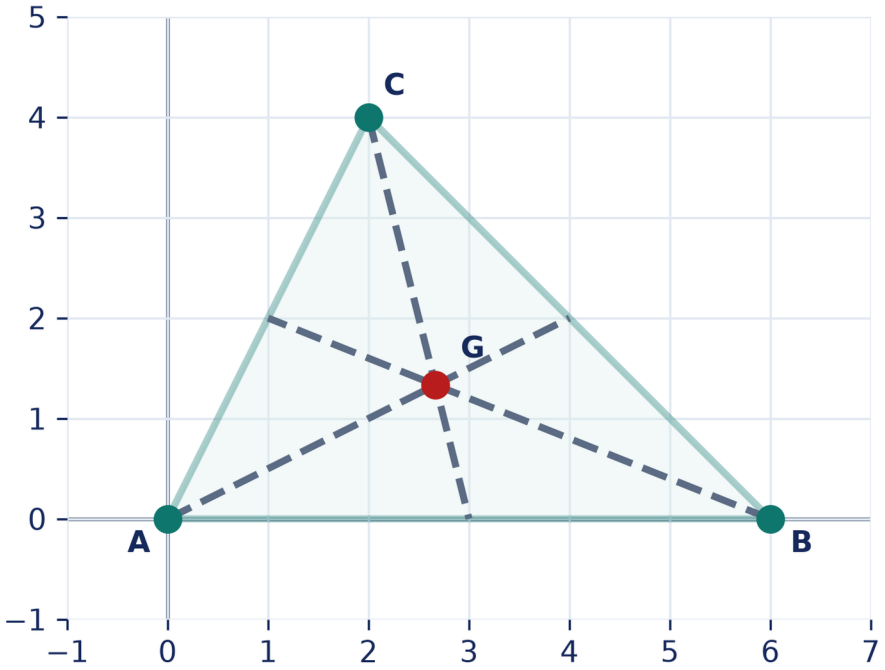
Centroids are equal-weight section formulas

For a triangle,

$$\mathbf{g} = \frac{\mathbf{a} + \mathbf{b} + \mathbf{c}}{3}.$$

For a tetrahedron with a fourth vertex $\mathbf{d}$, the centroid is $(\mathbf{a} + \mathbf{b} + \mathbf{c} + \mathbf{d})/4$. These are averages of position vectors, not averages of side lengths. To see the median concurrency, put A at the origin and let $D = (\mathbf{b} + \mathbf{c})/2$ be the midpoint of BC. Then $(2/3)D = (\mathbf{a} + \mathbf{b} + \mathbf{c})/3$ lies two-thirds along the median; symmetry gives the same point on all three medians.

The centroid lies two-thirds along every median



READ THE FIGURE

For A=(0,0), B=(6,0), C=(2,4), G=(8/3,4/3). On the median from A to (4,2), AG:GD=2:1.

LESSON 04 / 10

Two definitions describe the same scalar

For $\mathbf{a} = (a_1, a_2, a_3)$ and $\mathbf{b} = (b_1, b_2, b_3)$,

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 = |\mathbf{a}||\mathbf{b}|\cos\theta,$$

where $\theta \in [0, \pi]$ is the angle between two **nonzero** vectors. The dot product is commutative and distributes over addition, but its output is a scalar, not a vector.

SHORT DERIVATION

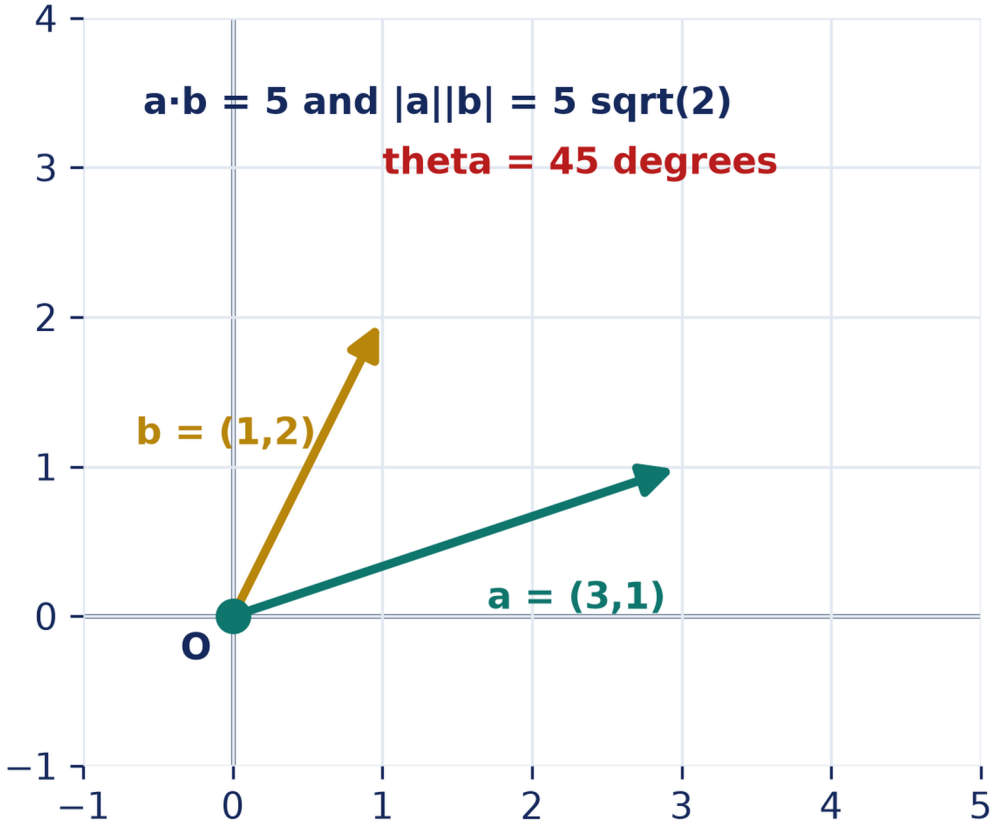
The cosine rule connects the two forms

The third side of the triangle on $\mathbf{a}, \mathbf{b}$ is $\mathbf{a} - \mathbf{b}$. Expand its squared length:

$$|\mathbf{a} - \mathbf{b}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2 - 2\mathbf{a} \cdot \mathbf{b}.$$

The cosine rule gives the same expression with $2|\mathbf{a}||\mathbf{b}|\cos\theta$ as the subtracted term. Compare coefficients to obtain the geometric dot product.

The dot product extracts the angle



READ THE FIGURE

cos(theta)=5/(sqrt(10) sqrt(5))=1/sqrt(2), so the smaller angle is 45 degrees.

Read geometry from the sign

For nonzero vectors:

- $\mathbf{a}\cdot\mathbf{b}=0$ *means perpendicular directions.*
- $\mathbf{a}\cdot\mathbf{b}>0$ *means an acute angle.*
- $\mathbf{a}\cdot\mathbf{b}<0$ *means an obtuse angle.*
- $\cos\theta=(\mathbf{a}\cdot\mathbf{b})/(|\mathbf{a}||\mathbf{b}|)$ *gives the actual angle.*

The zero vector has no angle. A statement such as "$\mathbf{a}\cdot\mathbf{b}=0$*, so the angle is* 90°*"* *silently needs* $\mathbf{a},\mathbf{b}\neq\mathbf{0}$.

WORKED EXAMPLE

An exact perpendicular pair and an exact angle

For $\mathbf{a}=(1,2,2)$ *and* $\mathbf{b}=(2,-2,1)$,

$$\mathbf{a}\cdot\mathbf{b}=2-4+2=0,\qquad |\mathbf{a}|=|\mathbf{b}|=3,$$

so the vectors are perpendicular. For $(3,1)$ *and* $(1,2)$ *in the figure, the same calculation gives* $\theta=45^\circ$ *without a diagram-based estimate.*

Cauchy–Schwarz is a one-line dot-product inequality

For $t\in\mathbb{R}$ *and* $\mathbf{b}\neq\mathbf{0}$,

$$0\le|\mathbf{a}-t\mathbf{b}|^2=|\mathbf{a}|^2-2t\mathbf{a}\cdot\mathbf{b}+t^2|\mathbf{b}|^2.$$

Choose $t=(\mathbf{a}\cdot\mathbf{b})/|\mathbf{b}|^2$ *to get*

$$(\mathbf{a}\cdot\mathbf{b})^2\le|\mathbf{a}|^2|\mathbf{b}|^2.$$

Equality holds exactly when the vectors are linearly dependent. This also explains why the cosine ratio lies between -1 *and* 1.

WORKED EXAMPLE

A numerical angle from components

For $\mathbf{a}=(1,2,3)$ *and* $\mathbf{b}=(4,5,6)$,

$$\mathbf{a}\cdot\mathbf{b}=32,\qquad |\mathbf{a}|^2=14,\qquad |\mathbf{b}|^2=77.$$

Therefore

$$\theta=\cos^{-1}\left(\frac{32}{\sqrt{1078}}\right)\approx12.93^\circ.$$

Cauchy–Schwarz is satisfied strictly: $32^2=1024<1078=14\cdot77$, *so the vectors are not parallel.*

LESSON 05 / 10

Projection keeps one component; rejection removes it

For $\mathbf{b} \neq \mathbf{0}$, the scalar projection of $\mathbf{a}$ along $\mathbf{b}$ and the vector projection are

$$\operatorname{scal}_{\mathbf{b}} \mathbf{a} = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{b}|}, \qquad \operatorname{proj}_{\mathbf{b}} \mathbf{a} = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{b}|^2}\mathbf{b}.$$

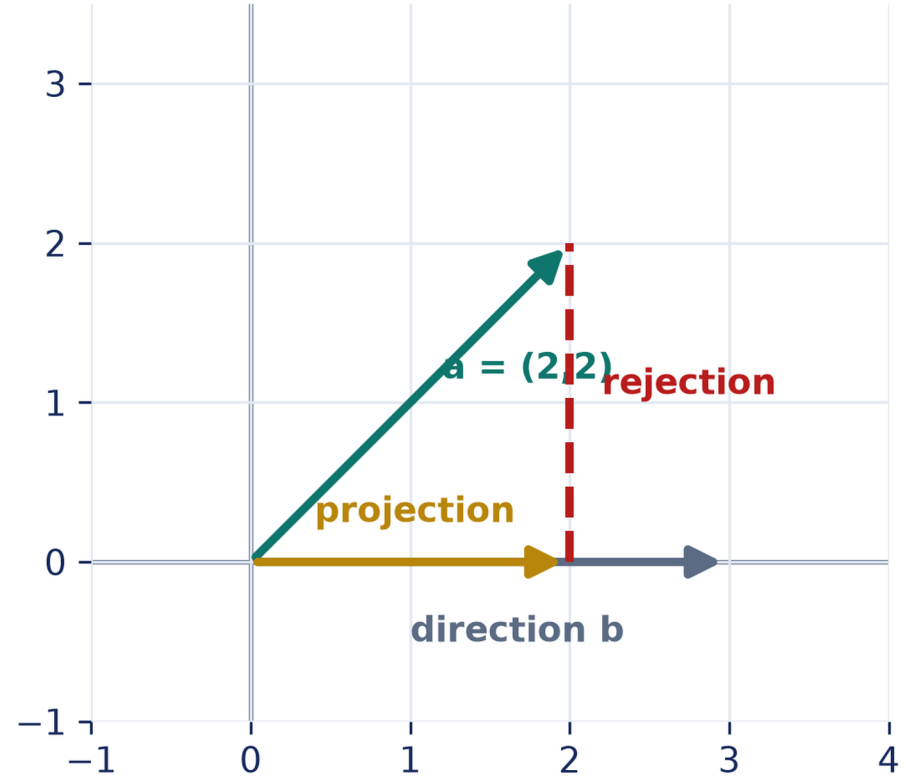
The scalar projection is signed. The vector projection points along $\mathbf{b}$ or $-\mathbf{b}$ according to that sign.

The perpendicular remainder, called the ***rejection***, is

$$\mathbf{a}_{\perp} = \mathbf{a} - \operatorname{proj}_{\mathbf{b}} \mathbf{a}, \qquad \mathbf{a}_{\perp}\cdot\mathbf{b} = 0.$$

Thus projection plus rejection reconstructs $\mathbf{a}$ exactly.

a = projection + perpendicular rejection



READ THE FIGURE
Along the x-direction, a=(2,2) has projection (2,0) and rejection (0,2). The two perpendicular pieces add back to a.

WORKED EXAMPLE

A 3D projection and rejection

Let $\mathbf{a} = (2, 2, 1)$ and $\mathbf{b} = (1, 2, 2)$. Since $\mathbf{a} \cdot \mathbf{b} = 8$ and $|\mathbf{b}|^2 = 9$,

$$\text{proj}_{\mathbf{b}}\,\mathbf{a} = \tfrac{8}{9}(1, 2, 2) = \left(\tfrac{8}{9}, \tfrac{16}{9}, \tfrac{16}{9}\right).$$

The scalar projection is $8/3$, and

$$\mathbf{a}_{\perp} = \left(\tfrac{10}{9}, \tfrac{2}{9}, -\tfrac{7}{9}\right).$$

A direct dot product with $(1, 2, 2)$ gives $10/9 + 4/9 - 14/9 = 0$.

Unit vectors construct angle bisectors

If $\mathbf{a}, \mathbf{b}$ are nonzero, then $\hat{\mathbf{a}} + \hat{\mathbf{b}}$ points along an internal angle bisector, provided it is nonzero. Their difference gives an external bisector direction. Normalizing first is essential: unequal lengths would otherwise pull the sum toward the longer vector.

WORKED EXAMPLE

An internal and an external bisector

For $\mathbf{a} = (3, 0, 0)$ and $\mathbf{b} = (0, 4, 0)$, the unit vectors are $(1, 0, 0)$ and $(0, 1, 0)$. Hence $(1, 1, 0)$ is an internal bisector direction and $(1, -1, 0)$ an external one. Their dot product is zero, as perpendicular internal and external bisectors require.

Work is a dot product; perpendicular force does none

For a constant force $\mathbf{F}$ acting through displacement $\mathbf{s}$, the work is

$$W = \mathbf{F} \cdot \mathbf{s} = |\mathbf{F}||\mathbf{s}|\cos\theta.$$

Only the component of force along displacement contributes. Negative work is meaningful: the force component opposes the displacement.

WORKED EXAMPLE

Work with exact components

For $\mathbf{F} = (3, 4, 5)$ and $\mathbf{s} = (1, 0, 2)$,

$$W = 3 \cdot 1 + 4 \cdot 0 + 5 \cdot 2 = 13.$$

The y-component of force contributes nothing because the displacement has no y-component.

COMMON TRAP

Projection onto b is not projection onto a

In general $\text{proj}_{\mathbf{b}}\,\mathbf{a} \neq \text{proj}_{\mathbf{a}}\,\mathbf{b}$. Also, scalar projection can be negative, while distance and magnitude cannot. Read the direction of "onto" before substituting.

LESSON 06 / 10

A cross product is a vector perpendicular to both inputs

For nonzero, nonparallel $\mathbf{a}, \mathbf{b}$, the vector $\mathbf{a} \times \mathbf{b}$ is perpendicular to both, has magnitude

$$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}|\sin\theta,$$

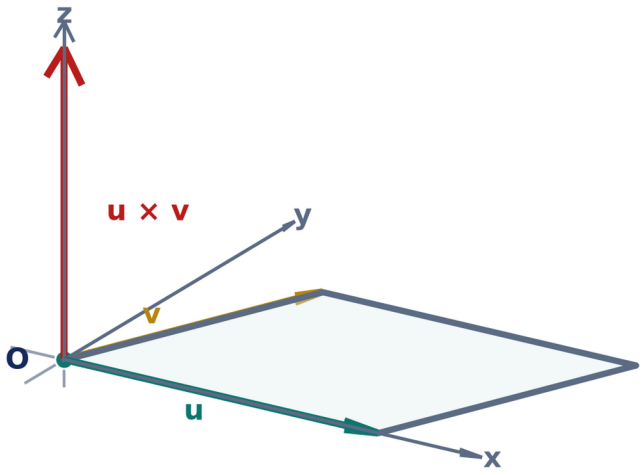
and follows the right-hand rule from $\mathbf{a}$ to $\mathbf{b}$. Its magnitude is the area of the parallelogram they span.

In components,

$$\mathbf{a} \times \mathbf{b} = (a_2b_3 - a_3b_2,\ a_3b_1 - a_1b_3,\ a_1b_2 - a_2b_1).$$

Notice the **minus sign in the middle component's expansion**, while the displayed component formula already includes the correct ordering.

The cross product is normal to the spanned plane



READ THE FIGURE

$u=(3,0,0)$, $v=(1,2,0)$, and $u\times v=(0,0,6)$. The shaded parallelogram has area 6, equal to the normal vector's length.

Order matters; parallelism gives zero

The cross product is anti-commutative and distributive:

$$\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}, \qquad \mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}.$$

For any vectors, $\mathbf{a} \times \mathbf{a} = \mathbf{0}$. Two nonzero vectors are parallel exactly when their cross product is $\mathbf{0}$. In general it is **not associative**; the vector triple product belongs to the extension lesson.

WORKED EXAMPLE

Compute and verify both perpendicularities

For $\mathbf{a} = (1, 2, 3)$ and $\mathbf{b} = (4, 5, 6)$,

$$\mathbf{a} \times \mathbf{b} = (-3, 6, -3), \qquad \mathbf{b} \times \mathbf{a} = (3, -6, 3).$$

Check:

$$(-3, 6, -3) \cdot (1, 2, 3) = 0, \qquad (-3, 6, -3) \cdot (4, 5, 6) = 0.$$

The area is $|(-3, 6, -3)| = 3\sqrt{6}$, and a unit normal is $(-1, 2, -1)/\sqrt{6}$.

Lagrange's identity joins dot and cross products

For every pair of vectors,

$$|\mathbf{a} \times \mathbf{b}|^2 + (\mathbf{a} \cdot \mathbf{b})^2 = |\mathbf{a}|^2 |\mathbf{b}|^2.$$

This is the algebraic form of $\sin^2\theta + \cos^2\theta = 1$. It can compute a cross-product magnitude without finding its components, or check an already computed pair.

WORKED EXAMPLE

A complete identity check

With $\mathbf{a} = (1, 2, 3)$ and $\mathbf{b} = (4, 5, 6)$, the previous example gives

$$|\mathbf{a} \times \mathbf{b}|^2 = 54, \qquad (\mathbf{a} \cdot \mathbf{b})^2 = 32^2 = 1024.$$

Therefore $54 + 1024 = 1078 = 14 \cdot 77$, exactly $|\mathbf{a}|^2 |\mathbf{b}|^2$.

For a second pair $\mathbf{u} = (3, -1, 2)$ and $\mathbf{v} = (1, 4, -2)$,

$$\mathbf{u} \times \mathbf{v} = (-6, 8, 13), \qquad \mathbf{u} \cdot \mathbf{v} = -5,$$

so $269 + 25 = 294 = 14 \cdot 21$. Thus $\cos\theta = -5/(7\sqrt{6})$ and $\sin^2\theta = 269/294$.

COMMON TRAP

Area is nonnegative; orientation is not

Use $|\mathbf{a} \times \mathbf{b}|$ for area. The vector itself points along one of two opposite normals, determined by factor order. Reversing the order preserves area but reverses oriented area and normal direction.

LESSON 07 / 10

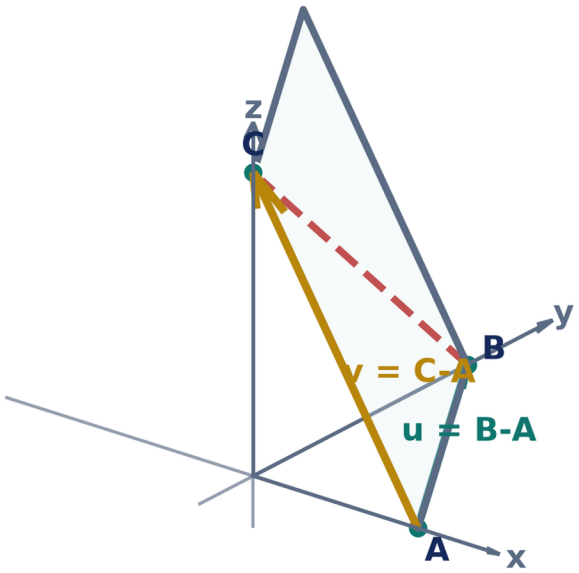
Use two edges from the same vertex

For a parallelogram spanned by $\mathbf{u}, \mathbf{v}$ and a triangle with the same two sides,

$$[\![\text{parallelogram}]\!] = |\mathbf{u} \times \mathbf{v}|, \qquad [\![\text{triangle}]\!] = \tfrac{1}{2}|\mathbf{u} \times \mathbf{v}|.$$

For $\triangle ABC$, take $\mathbf{u} = \mathbf{b} - \mathbf{a}$ and $\mathbf{v} = \mathbf{c} - \mathbf{a}$. Do not use the three position vectors directly unless A is the origin.

The diagonal halves the edge parallelogram



READ THE FIGURE

u=B-A=(-1,2,0), v=C-A=(-1,0,3), and u×v=(6,3,2). Parallelogram area 7 gives triangle area 7/2.

WORKED EXAMPLE

An exact 3D triangle area

For $A = (1, 0, 0)$, $B = (0, 2, 0)$ and $C = (0, 0, 3)$,

$$\mathbf{u} = (-1, 2, 0), \qquad \mathbf{v} = (-1, 0, 3),$$

$$\mathbf{u} \times \mathbf{v} = (6, 3, 2), \qquad |\mathbf{u} \times \mathbf{v}| = \sqrt{36 + 9 + 4} = 7.$$

Hence the triangle area is $7/2$ and the parallelogram area is 7. The oriented vector area of the ordered triangle is $(6, 3, 2)/2 = (3, 3/2, 1)$.

Cross products prove area ratios without coordinates

If two triangles share the same base and their third vertices lie on a line parallel to that base, their areas are equal: the cross-product magnitude retains the same perpendicular height. A median divides a triangle into two equal-area triangles because the two new bases are equal and the height is shared.

For a parallelogram on $\mathbf{a}, \mathbf{b}$, its diagonals are $\mathbf{a}+\mathbf{b}$ and $\mathbf{a}-\mathbf{b}$. Their squared lengths sum to $2(|\mathbf{a}|^2+|\mathbf{b}|^2)$, while

$$(\mathbf{a}+\mathbf{b})\times(\mathbf{a}-\mathbf{b}) = -2(\mathbf{a}\times\mathbf{b}).$$

This gives the area of the parallelogram formed by the diagonals as twice the original parallelogram area.

The cosine rule and area formula are companion identities

With $c=|\mathbf{a}-\mathbf{b}|$, expansion gives

$$c^2 = |\mathbf{a}|^2+|\mathbf{b}|^2-2\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}|^2+|\mathbf{b}|^2-2|\mathbf{a}||\mathbf{b}|\cos\theta.$$

At the same time, the triangle area is

$$\tfrac{1}{2}|\mathbf{a}\times\mathbf{b}| = \tfrac{1}{2}|\mathbf{a}||\mathbf{b}|\sin\theta.$$

Thus one expansion handles obtuse angles automatically, and the other handles area without drawing a perpendicular height.

SHORT DERIVATION

Angle bisector theorem from section weights

In $\triangle ABC$, let the internal bisector from A meet BC at D. Unit directions along AB and AC add to give the bisector direction. Intersecting that direction with BC yields

$$D = \frac{|AC|B+|AB|C}{|AB|+|AC|}.$$

Compare this with the section formula to obtain $BD:DC = AB:AC$. The vector proof keeps the side lengths attached to the correct opposite weights.

KEY RESULT

A fast area self-check

Before accepting a cross-product answer, verify that it is perpendicular to both edge vectors and that its length matches a rough base-times-height estimate. These two checks catch most component-order and sign errors.

LESSON 08 / 10

A signed volume in one scalar

The **scalar triple product** is

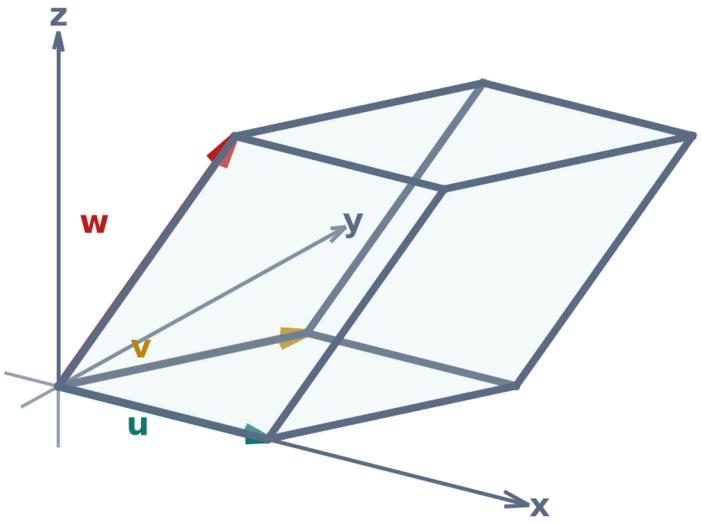
$$[\mathbf{a}\ \mathbf{b}\ \mathbf{c}] = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}).$$

First $\mathbf{b} \times \mathbf{c}$ gives the oriented base area; the dot product with $\mathbf{a}$ then multiplies its magnitude by the signed height. Consequently,

$$V_{\text{parallelepiped}} = |[\mathbf{a}\ \mathbf{b}\ \mathbf{c}]|.$$

The sign records orientation, not a negative physical volume.

Volume = base area × perpendicular height



READ THE FIGURE
For u=(2,0,0), v=(1,2,0), w=(1,1,2), the triple product is 8. The base area is 4 and the perpendicular height is 2.

Permutations follow determinant rules

Cyclic permutations preserve the scalar triple product, while swapping two factors reverses its sign:

$$[\mathbf{a}\ \mathbf{b}\ \mathbf{c}] = [\mathbf{b}\ \mathbf{c}\ \mathbf{a}] = [\mathbf{c}\ \mathbf{a}\ \mathbf{b}],$$

$$[\mathbf{a}\ \mathbf{b}\ \mathbf{c}] = -[\mathbf{b}\ \mathbf{a}\ \mathbf{c}].$$

Also $[\mathbf{a}\ \mathbf{b}\ \mathbf{c}] = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$. If any two factors are equal, the triple product is zero.

WORKED EXAMPLE

Compute once, then use permutations

For $\mathbf{a} = (1, 2, 3)$, $\mathbf{b} = (4, 5, 6)$ and $\mathbf{c} = (-1, 0, 2)$,

$$\mathbf{b} \times \mathbf{c} = (10, -14, 5), \qquad [\mathbf{a}\ \mathbf{b}\ \mathbf{c}] = 10 - 28 + 15 = -3.$$

Thus the parallelepiped volume is 3, while $[\mathbf{b}\ \mathbf{a}\ \mathbf{c}] = +3$ and $[\mathbf{b}\ \mathbf{c}\ \mathbf{a}] = -3$.

Zero is the coplanarity/dependence test

Three vectors are linearly dependent exactly when their scalar triple product is zero. Geometrically, their spans have no nonzero three-dimensional volume. This includes coplanar position vectors through the origin, parallel factors, and the zero vector.

For four **points** A, B, C, D, first form edge vectors from one vertex:

$$[\mathbf{b} - \mathbf{a}\ \mathbf{c} - \mathbf{a}\ \mathbf{d} - \mathbf{a}] = 0 \iff A, B, C, D \text{ are coplanar.}$$

If the first two edges are parallel, the four points can be collinear or otherwise degenerate; zero still correctly signals dependence, but not necessarily a unique plane.

WORKED EXAMPLE

A four-point test with exact edges

Test $A = (1, 2, 3)$, $B = (4, 5, 6)$, $C = (-1, 0, 2)$, $D = (2, -1, 4)$. From A,

$$\mathbf{u} = (3, 3, 3), \quad \mathbf{v} = (-2, -2, -1), \quad \mathbf{w} = (1, -3, 1).$$

Then $\mathbf{v} \times \mathbf{w} = (-5, 1, 8)$ and

$$[\mathbf{u}\ \mathbf{v}\ \mathbf{w}] = 3(-5) + 3(1) + 3(8) = 12 \neq 0.$$

The four points are not coplanar; their edge parallelepiped has volume 12 and their tetrahedron has volume 2.

COMMON TRAP

Signed triple product versus volume

Never report a negative volume. Keep the sign while proving permutation or orientation identities, then take the absolute value only when the question asks for volume. For a tetrahedron, also apply the factor $1/6$.

LESSON 09 / 10

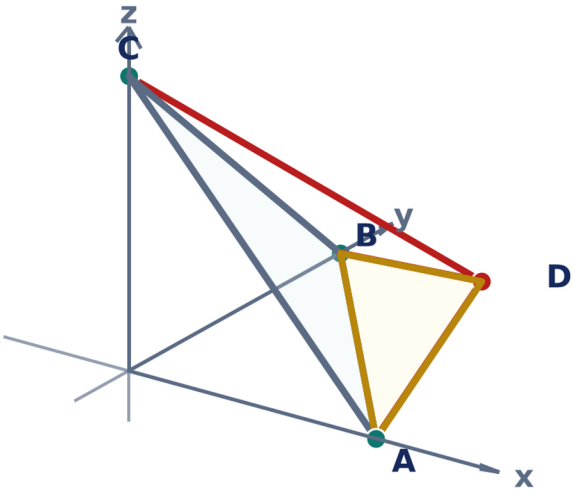
A tetrahedron is one-sixth of an edge parallelepiped

For vertices A, B, C, D, set $\mathbf{u} = \mathbf{b} - \mathbf{a}$, $\mathbf{v} = \mathbf{c} - \mathbf{a}$, $\mathbf{w} = \mathbf{d} - \mathbf{a}$. Then

$$V_{\text{tetra}} = \frac{1}{6}|[\mathbf{u}\ \mathbf{v}\ \mathbf{w}]|, \qquad \mathbf{g} = \frac{\mathbf{a}+\mathbf{b}+\mathbf{c}+\mathbf{d}}{4}.$$

The factor is $1/6$, not $1/3$: the triangular pyramid on a fixed parallelepiped base has one-third of the corresponding triangular-prism volume, and that prism is half the parallelepiped.

Use three edges from one tetrahedron vertex



READ THE FIGURE
Taking D as the common tail gives edges A-D, B-D, C-D. Their scalar triple product has absolute value 5, so the tetrahedron volume is 5/6.

WORKED EXAMPLE

Volume and centroid from four exact vertices

For $A = (1, 0, 0)$, $B = (0, 2, 0)$, $C = (0, 0, 3)$ and $D = (1, 1, 1)$, use D as the common tail:

$$\mathbf{u} = A - D = (0, -1, -1), \quad \mathbf{v} = B - D = (-1, 1, -1), \quad \mathbf{w} = C - D = (-1, -1, 2).$$

Then $\mathbf{v} \times \mathbf{w} = (1, 3, 2)$ and $[\mathbf{u}\ \mathbf{v}\ \mathbf{w}] = -5$. Therefore

$$V = \frac{|-5|}{6} = \frac{5}{6}, \qquad G = \frac{A+B+C+D}{4} = \left(\frac{1}{2}, \frac{3}{4}, 1\right).$$

A parameter problem can be one determinant

Suppose $\mathbf{a} = (1, 2, 3)$, $\mathbf{b} = (4, 5, 6)$ and $\mathbf{c} = (-1, 0, \lambda)$ are coplanar through the origin. Compute

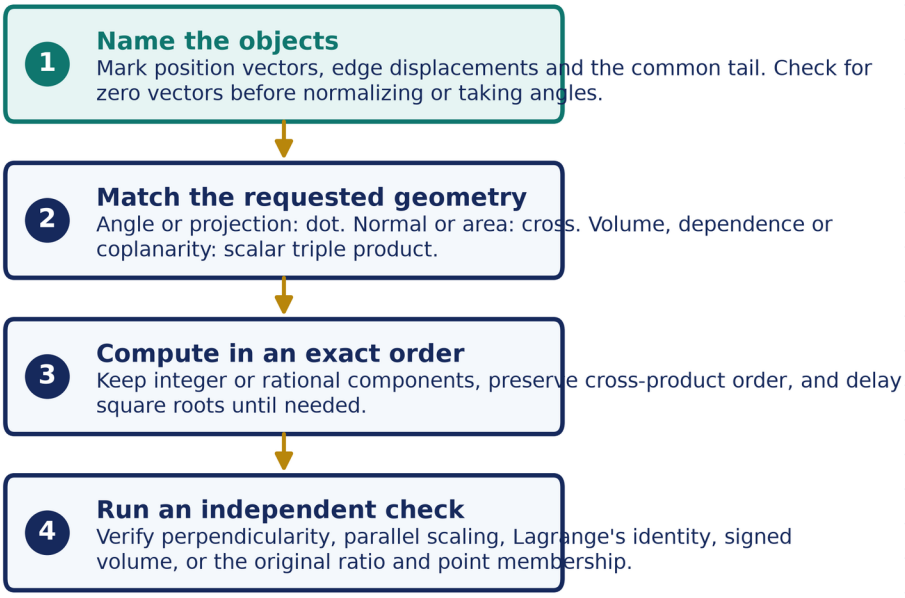
$$\mathbf{b} \times \mathbf{c} = (5\lambda,\ -6 - 4\lambda, 5),$$

$$[\mathbf{a}\ \mathbf{b}\ \mathbf{c}] = 5\lambda + 2(-6 - 4\lambda) + 15 = 3(1 - \lambda).$$

Coplanarity forces $\lambda = 1$. At $\lambda = 2$, the triple product is -3, matching the nonzero-volume example in the previous lesson.

Choose the smallest sufficient computation

A reliable vector-problem workflow



READ THE FIGURE

The product is chosen by the geometry. A second identity or direct substitution should confirm the result.

WORKED EXAMPLE

A mixed checklist in action

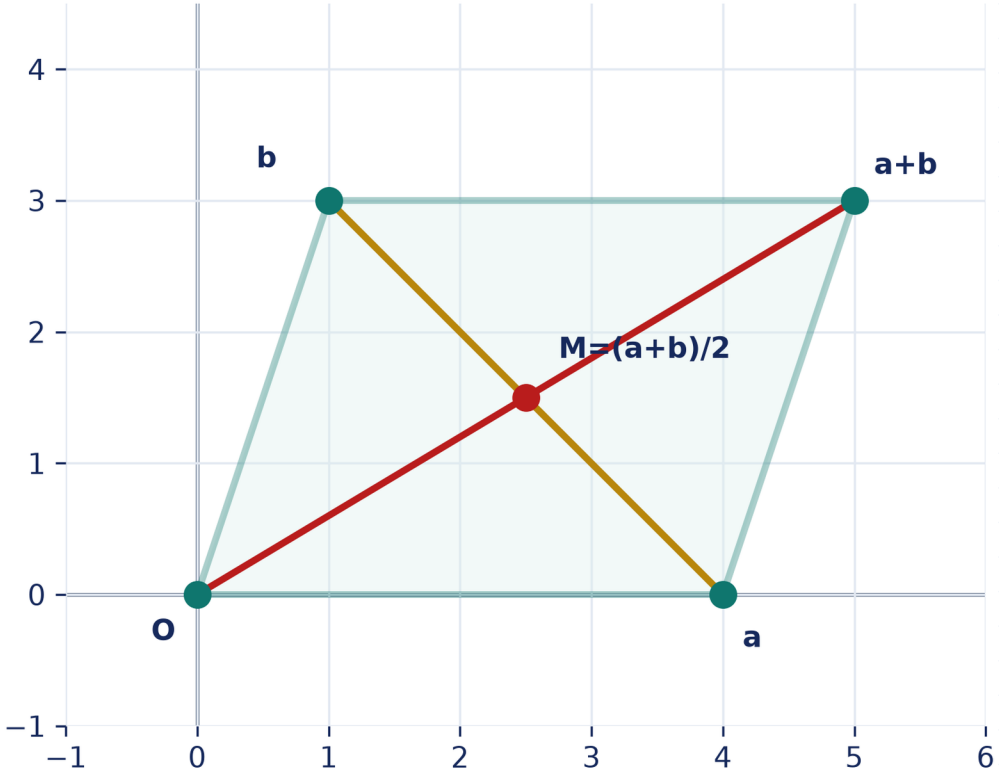
If a question asks for a tetrahedron's volume and whether a fourth point lies in a base plane, use the ***same three edge vectors*** for both. A zero triple product answers coplanarity; otherwise $|[\mathbf{u}\ \mathbf{v}\ \mathbf{w}]|/6$ answers volume. There is no need to compute a plane equation for the vector unit.

LESSON 10 / 10

Translate the origin to expose symmetry

For a parallelogram with vertices $0, \mathbf{a}, \mathbf{a}+\mathbf{b}, \mathbf{b}$, both diagonals have midpoint $(\mathbf{a}+\mathbf{b})/2$. Placing one vertex at the origin turns a synthetic bisection statement into one average.

Both diagonals have the same midpoint



READ THE FIGURE

For a=(4,0), b=(1,3), M=(5/2,3/2). It is the midpoint of both 0 to a+b and a to b.

Visual check: $\overrightarrow{OM} = (5/2, 3/2)$, while the other diagonal has endpoints $(4, 0)$ and $(1, 3)$, whose average is identical. No length formula is needed.

KEY RESULT

Second read: the two diagonals also obey the parallelogram law

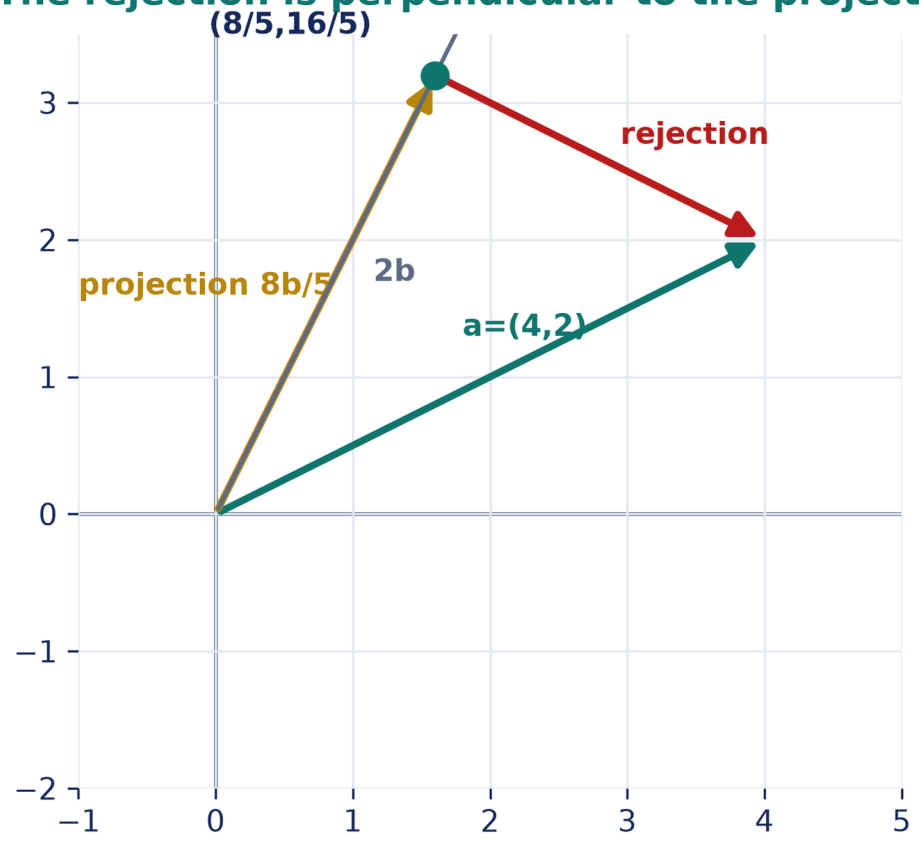
Here $\mathbf{a}+\mathbf{b} = (5, 3)$ and $\mathbf{a}-\mathbf{b} = (3, -3)$, so the squared diagonal lengths are 34 and 18. Their sum is $52 = 2(16+10)$, exactly twice the sum of the squared side lengths.

Projection and rejection make a right triangle

For $\mathbf{a}=(4,2)$ and $\mathbf{b}=(1,2)$, we have $\mathbf{a}\cdot\mathbf{b}=8$ and $|\mathbf{b}|^2=5$. Therefore

$$\operatorname{proj}_{\mathbf{b}}\mathbf{a}=\tfrac{8}{5}(1,2), \qquad \mathbf{a}_{\perp}=\left(\tfrac{12}{5},-\tfrac{6}{5}\right).$$

The rejection is perpendicular to the projection



READ THE FIGURE

Projection plus rejection is (8/5+12/5,16/5-6/5)=(4,2). The rejection dots (1,2) to zero.

Visual check: $(12/5,\,-6/5)\cdot(1,2)=12/5-12/5=0$, and the scalar projection is $8/\sqrt{5}$.

KEY RESULT

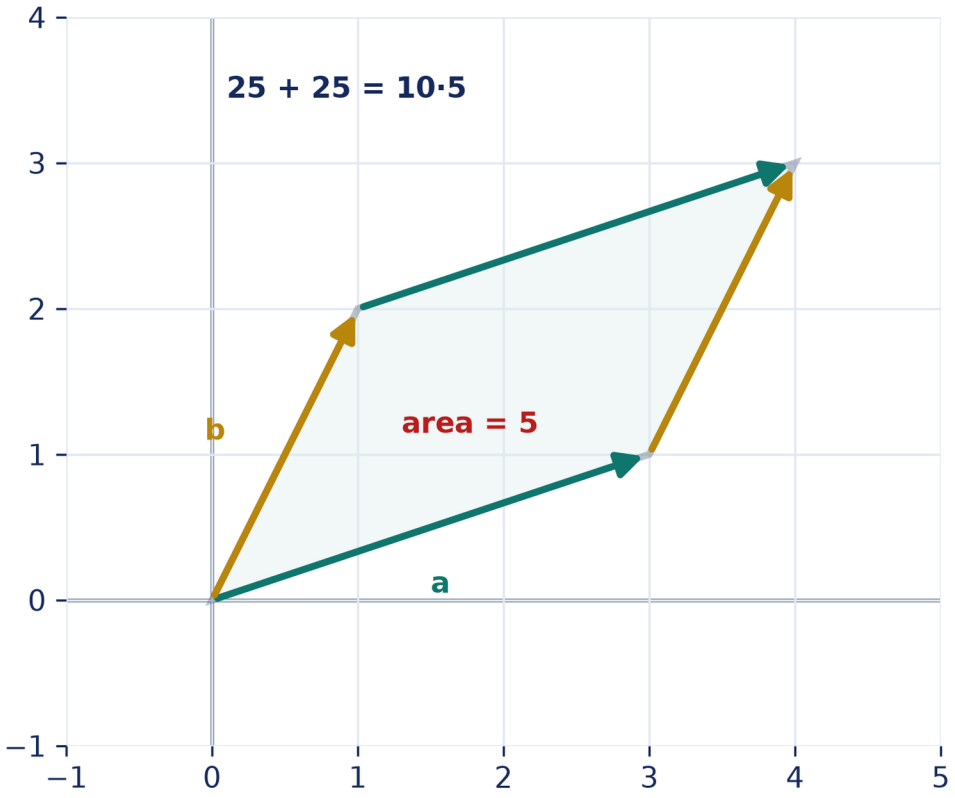
Second read: Pythagoras checks the decomposition

The squared projection length is $(64/25)|(1,2)|^2=64/5$, and the squared rejection length is $(144+36)/25=36/5$. They sum to $100/5=20=|(4,2)|^2$, while the rejection's length is $6/\sqrt{5}$.

One parallelogram checks dot, cross and angle

Take $\mathbf{a} = (3, 1)$ and $\mathbf{b} = (1, 2)$. Then $\mathbf{a} \cdot \mathbf{b} = 5$ and the z-component of $\mathbf{a} \times \mathbf{b}$ is $3 \cdot 2 - 1 \cdot 1 = 5$.

Area 5 and angle 45 degrees in one construction



READ THE FIGURE

Cross magnitude squared 25 plus dot squared 25 equals |a|^2|b|^2=50. Also cos(theta)=1/sqrt(2).

Visual check: the parallelogram area is 5, the triangle on the same edges is $5/2$, and Lagrange's identity reads $25 + 25 = 50$.

KEY RESULT

Second read: the diagonals span twice the area

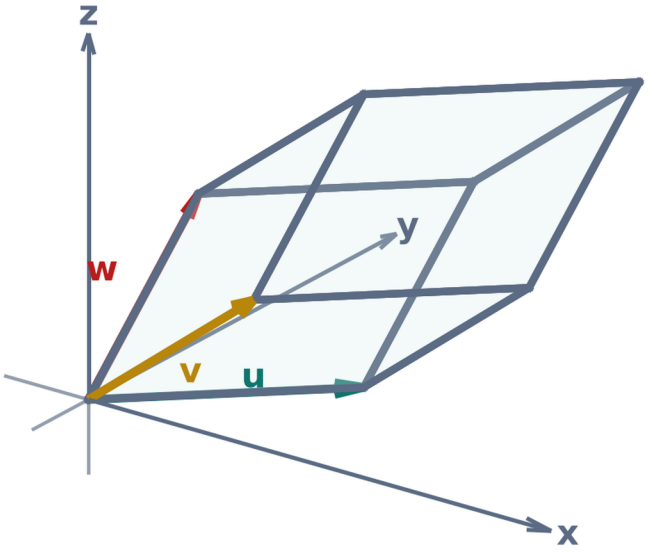
The diagonals are $\mathbf{a} + \mathbf{b} = (4, 3)$ and $\mathbf{a} - \mathbf{b} = (2, -1)$. Their planar cross determinant is $4(-1) - 3(2) = -10$, whose absolute value is twice the original area 5, as the diagonal identity predicts.

A triple product splits into base and height

For $\mathbf{u}=(1,1,0)$, $\mathbf{v}=(1,0,1)$ and $\mathbf{w}=(0,1,1)$,

$$\mathbf{v}\times\mathbf{w}=(-1,\,-1,\,1),\qquad [\mathbf{u}\;\mathbf{v}\;\mathbf{w}]=-2.$$

Signed triple product -2 means volume 2



READ THE FIGURE

The base spanned by v,w has area sqrt(3), so the height is 2/sqrt(3). The corresponding tetrahedron has volume 1/3.

Visual check: $|[\mathbf{u}\;\mathbf{v}\;\mathbf{w}]|=2$; swapping any two edges changes the sign but not the volume.

KEY RESULT

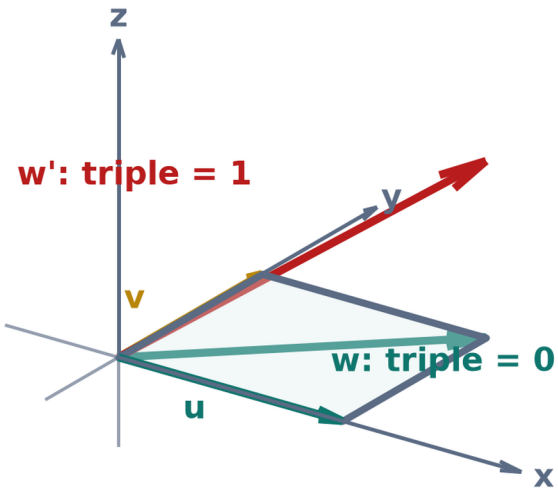
Second read: the tetrahedron uses the same three edges

The tetrahedron with vertices $\mathbf{0},\mathbf{u},\mathbf{v},\mathbf{w}$ has volume $2/6=1/3$. Its centroid is $(\mathbf{u}+\mathbf{v}+\mathbf{w})/4=(1/2,1/2,1/2)$, so volume and centroid need no new edge computation.

Zero volume is visible dependence

Keep the base edges $\mathbf{u} = (1, 0, 0)$ and $\mathbf{v} = (0, 1, 0)$ fixed. For $D = (1, 1, 0)$, the edge $\mathbf{w} = (1, 1, 0)$ lies in the base plane. For $D' = (1, 1, 1)$, the edge $\mathbf{w}' = (1, 1, 1)$ lifts out of it.

Coplanar edges collapse the volume to zero



READ THE FIGURE

With u=i and v=j, the triple product is the z-component of w. It is 0 for w=(1,1,0) and 1 for w'=(1,1,1).

Visual check: the zero case does not imply that every edge is zero; it implies that the three edges fail to span a three-dimensional volume.

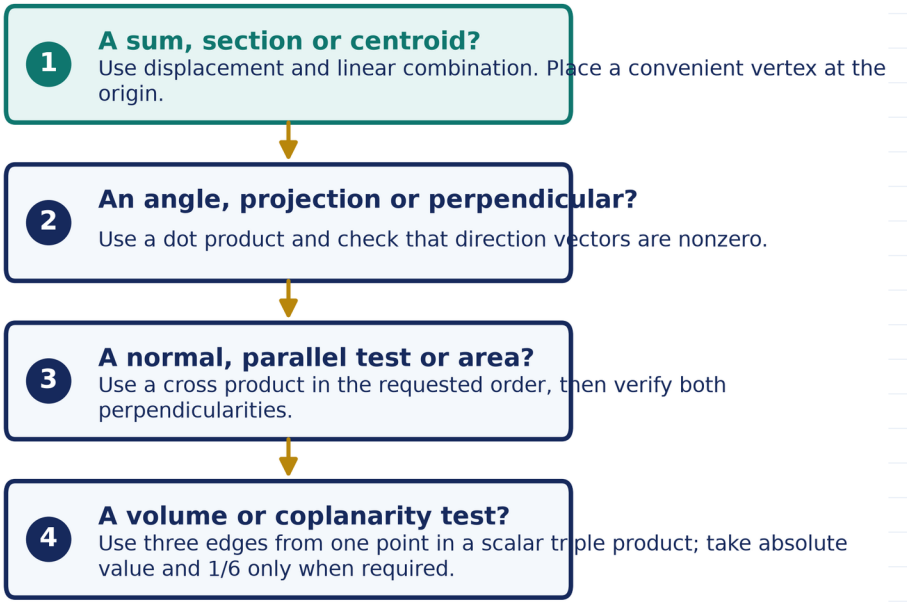
KEY RESULT

Second read: height is the remaining coordinate

For a general third edge $\mathbf{w} = (s, t, h)$ with $\mathbf{u} = \mathbf{i}$ and $\mathbf{v} = \mathbf{j}$, the triple product is simply h. Every $h = 0$ point lies in the base plane; for $h \neq 0$, the parallelepiped volume is $|h|$ because the base is a unit square.

Let the picture choose the product

Read the geometry before doing component arithmetic



READ THE FIGURE

A sketch identifies the right operation and supplies an independent geometric estimate; exact components supply the proof.

The three products answer different questions

Operation	Output	Zero condition	Geometric meaning
a·b	scalar	perpendicular nonzero vectors	angle and projection
a×b	vector	parallel or zero input	normal and area
a·(b×c)	scalar	dependent/coplanar edges	signed volume

READ THE FIGURE

Do not interchange the zero tests: dot zero is perpendicularity, whereas cross zero is parallelism or a zero vector.

KEY RESULT

The final visual habit

After computing, return to the sketch. Is every displacement head minus tail? Are direction vectors nonzero? Does the projection point the right way and the normal lie on the expected side? Is area positive, and did a tetrahedron include 1/6? These visual and structural checks catch errors that repeated algebra in the same mistaken setup cannot.

LESSON 01 / 03

The vector triple product returns to a plane

The **BAC–CAB** identity is

$$\mathbf{a}\times(\mathbf{b}\times\mathbf{c})=\mathbf{b}(\mathbf{a}\cdot\mathbf{c})-\mathbf{c}(\mathbf{a}\cdot\mathbf{b}).$$

The result is a linear combination of $\mathbf{b},\mathbf{c}$, as it must be: a vector perpendicular to $\mathbf{b}\times\mathbf{c}$ lies in their span. The identity can be proved by linearity and checking basis vectors, or by expanding components once.

The other grouping gives

$$(\mathbf{a}\times\mathbf{b})\times\mathbf{c}=\mathbf{b}(\mathbf{a}\cdot\mathbf{c})-\mathbf{a}(\mathbf{b}\cdot\mathbf{c}).$$

The two expressions are different in general. Cross products need parentheses.

WORKED EXAMPLE

Two groupings, two different answers

For $\mathbf{a}=(7,8,9)$, $\mathbf{b}=(1,2,3)$ and $\mathbf{c}=(4,5,6)$,

$$\mathbf{a}\cdot\mathbf{c}=122,\qquad \mathbf{a}\cdot\mathbf{b}=50,\qquad \mathbf{b}\cdot\mathbf{c}=32.$$

Hence

$$\mathbf{a}\times(\mathbf{b}\times\mathbf{c})=122\mathbf{b}-50\mathbf{c}=(-78,\,-6,66),$$

whereas

$$(\mathbf{a}\times\mathbf{b})\times\mathbf{c}=122\mathbf{b}-32\mathbf{a}=(-102,\,-12,78).$$

They are unequal, but each expansion agrees with direct cross-product computation.

COMMON TRAP

Not associative does not mean never equal

The two groupings may coincide for special dependent or orthogonal vectors. "Not associative" means no universal equality is available. Prove a special case if you want to remove parentheses; never remove them by habit.

Jacobi's identity is three expansions cancelling

For every $\mathbf{a},\mathbf{b},\mathbf{c}$,

$$\mathbf{a}\times(\mathbf{b}\times\mathbf{c})+\mathbf{b}\times(\mathbf{c}\times\mathbf{a})+\mathbf{c}\times(\mathbf{a}\times\mathbf{b})=\mathbf{0}.$$

SHORT DERIVATION

Cancellation by cyclic terms

Apply BAC–CAB to each term:

$$\mathbf{b}(\mathbf{a}\cdot\mathbf{c})-\mathbf{c}(\mathbf{a}\cdot\mathbf{b})+\mathbf{c}(\mathbf{b}\cdot\mathbf{a})-\mathbf{a}(\mathbf{b}\cdot\mathbf{c})+\mathbf{a}(\mathbf{c}\cdot\mathbf{b})-\mathbf{b}(\mathbf{c}\cdot\mathbf{a}).$$

Commutativity of the dot product makes corresponding vector coefficients cancel in pairs. The numerical example above therefore gives a zero cyclic sum even though neither individual triple product is zero.

A scalar quadruple product generalizes Lagrange

Another useful identity is

$$(\mathbf{a}\times\mathbf{b})\cdot(\mathbf{c}\times\mathbf{d})=(\mathbf{a}\cdot\mathbf{c})(\mathbf{b}\cdot\mathbf{d})-(\mathbf{a}\cdot\mathbf{d})(\mathbf{b}\cdot\mathbf{c}).$$

Setting $\mathbf{c}=\mathbf{a}$ and $\mathbf{d}=\mathbf{b}$ recovers Lagrange's identity. This form is valuable when two planes or two oriented areas must be compared without computing every cross component separately.

Keep the triple-product shapes distinct

Expression	Type	Main interpretation
a·(b×c)	scalar	signed volume / dependence
a×(b×c)	vector	BAC–CAB in span(b,c)
(a×b)·(c×d)	scalar	dot of two oriented normals
cyclic Jacobi sum	zero vector	three expansions cancel

READ THE FIGURE

First identify whether the answer is a scalar or a vector. That simple type check rejects many incorrect identities immediately.

WORKED EXAMPLE

A quick scalar-quadruple check

Let $\mathbf{a}=\mathbf{i}$, $\mathbf{b}=\mathbf{j}$, $\mathbf{c}=\mathbf{i}+\mathbf{j}$ and $\mathbf{d}=\mathbf{j}+\mathbf{k}$. Then

$$(\mathbf{a}\times\mathbf{b})\cdot(\mathbf{c}\times\mathbf{d})=\mathbf{k}\cdot(\mathbf{i}-\mathbf{j}+\mathbf{k})=1.$$

The right side is $(\mathbf{a}\cdot\mathbf{c})(\mathbf{b}\cdot\mathbf{d})-(\mathbf{a}\cdot\mathbf{d})(\mathbf{b}\cdot\mathbf{c})=1\cdot1-0\cdot1=1$.

LESSON 02 / 03

Barycentric coordinates are vector weights and area ratios

For a nondegenerate triangle ABC, normalized barycentric coordinates write

$$P = \alpha A + \beta B + \gamma C, \qquad \alpha + \beta + \gamma = 1.$$

For a point inside the triangle,

$$\alpha = \frac{[PBC]}{[ABC]}, \qquad \beta = \frac{[PCA]}{[ABC]}, \qquad \gamma = \frac{[PAB]}{[ABC]}.$$

Outside the triangle, use signed areas and signed weights. Unnormalized coordinates $(\lambda_1 : \lambda_2 : \lambda_3)$ become normalized after division by their nonzero sum.

WORKED EXAMPLE

Coordinates from areas and back again

For $A = (0, 0)$, $B = (4, 0)$, $C = (0, 3)$ and $P = (1, 1)$,

$$P = \frac{5}{12}A + \frac{1}{4}B + \frac{1}{3}C.$$

Thus the unnormalized coordinates are $(5 : 3 : 4)$. Also $[ABC] = 6$ and $[PBC] = 5/2$, so $[PBC]/[ABC] = 5/12$, matching the coefficient of A. The three subtriangle area ratios are $5/12, 1/4, 1/3$, whose sum is 1.

Ceva is a product of directed ratios

Put

$$x = \frac{BX}{XC}, \qquad y = \frac{CY}{YA}, \qquad z = \frac{AZ}{ZB},$$

where $X \in BC$, $Y \in CA$ and $Z \in AB$. The cevians AX, BY, CZ are concurrent exactly when

$$xyz = 1,$$

with directed ratios used for external points. In barycentrics, $X = (0 : 1 : x)$, $Y = (y : 0 : 1)$ and $Z = (1 : z : 0)$; eliminating the cevian equations gives the same product condition.

WORKED EXAMPLE

A concurrent Ceva configuration

If $BX : XC = 2 : 1$, $CY : YA = 3 : 1$ and $AZ : ZB = 1 : 6$, then

$$x = 2, \qquad y = 3, \qquad z = \frac{1}{6}, \qquad xyz = 1.$$

Therefore AX, BY, CZ are concurrent. No coordinates of the intersection are required for the conclusion.

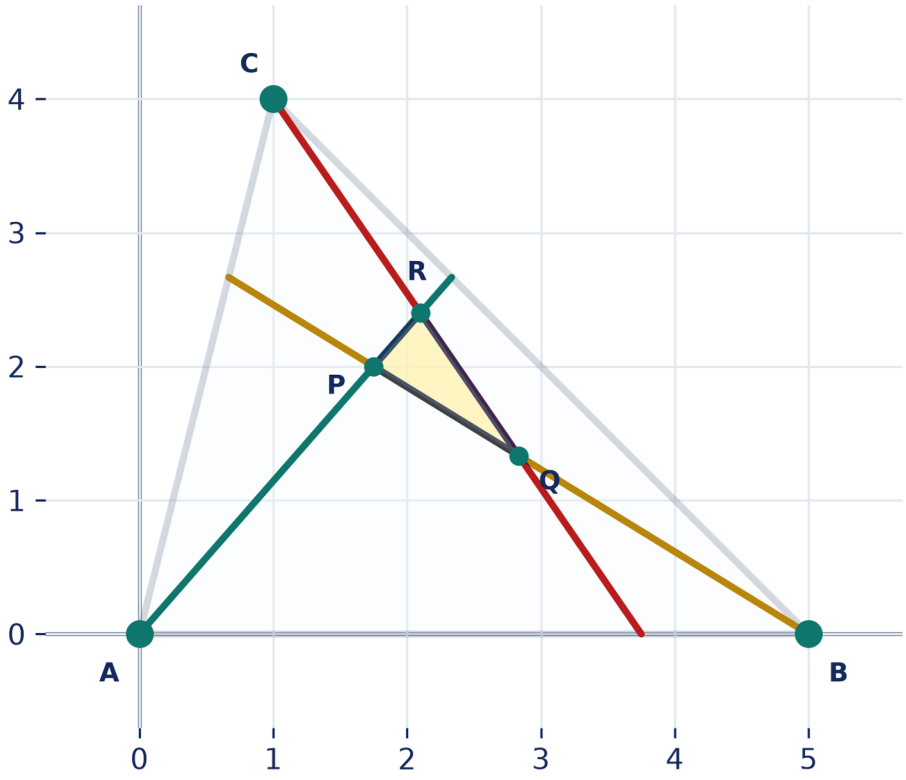
Routh measures the triangle bounded by three cevians

With the same ratio convention and $xyz \neq 1$, the cevians bound a triangle PQR satisfying

$$\frac{[PQR]}{[ABC]} = \frac{(xyz-1)^2}{(xy+x+1)(yz+y+1)(zx+z+1)}.$$

When $xyz = 1$, the cevians concur and the numerator correctly becomes zero.

The three cevians bound the central triangle PQR



READ THE FIGURE
A=(0,0), B=(5,0), C=(1,4), x=2, y=1/2, z=3. Direct cevian intersections give P=(7/4,2), Q=(17/6,4/3), R=(21/10,12/5).

WORKED EXAMPLE

An exact Routh calculation, checked by intersections

For $x = 2$, $y = 1/2$ and $z = 3$, we have $xyz = 3$ and

$$xy + x + 1 = 4, \qquad yz + y + 1 = 3, \qquad zx + z + 1 = 10.$$

Therefore

$$\frac{[PQR]}{[ABC]} = \frac{(3-1)^2}{4 \cdot 3 \cdot 10} = \frac{1}{30}.$$

The displayed coordinates independently give the same ratio. Since $[ABC] = 10$, the central triangle has area $1/3$. Keep the cyclic factors in order: here $yz + y + 1 = 3/2 + 1/2 + 1 = 3$, not 2; using 2 would give the incorrect result $1/20$.

LESSON 03 / 03

Choose an origin that makes the theorem disappear

Vector proofs become short when the origin matches the symmetry. For a parallelogram, use vertices $0, \mathbf{a}, \mathbf{a}+\mathbf{b}, \mathbf{b}$. For a triangle's medians, put one vertex at the origin. For a circumcentre problem, put the circumcentre at the origin.

WORKED EXAMPLE

Varignon's parallelogram in one subtraction

Let A, B, C, D be any four points in a plane, and let P, Q, R, S be the midpoints of AB, BC, CD, DA. Then

$$Q - P = \frac{B + C - A - B}{2} = \frac{C - A}{2},$$

$$R - S = \frac{C + D - D - A}{2} = \frac{C - A}{2}.$$

Thus $PQ \parallel SR$ and $PQ = SR$; similarly $QR \parallel PS$. The midpoint quadrilateral is always a parallelogram, even when the original quadrilateral is not one.

With the circumcentre at O = 0, H = A+B+C

Suppose the circumcentre O of $\triangle ABC$ is the origin, so $|A| = |B| = |C| = R$. Define $H = A + B + C$. Then

$$(H - A) \cdot (C - B) = (B + C) \cdot (C - B) = |C|^2 - |B|^2 = 0.$$

Thus $AH \perp BC$; cyclic relabelling gives the other two altitudes. Hence H is the orthocentre.

The centroid is $G = (A + B + C)/3 = H/3$. Therefore O, G, H are collinear and

$$OG : GH = 1 : 2.$$

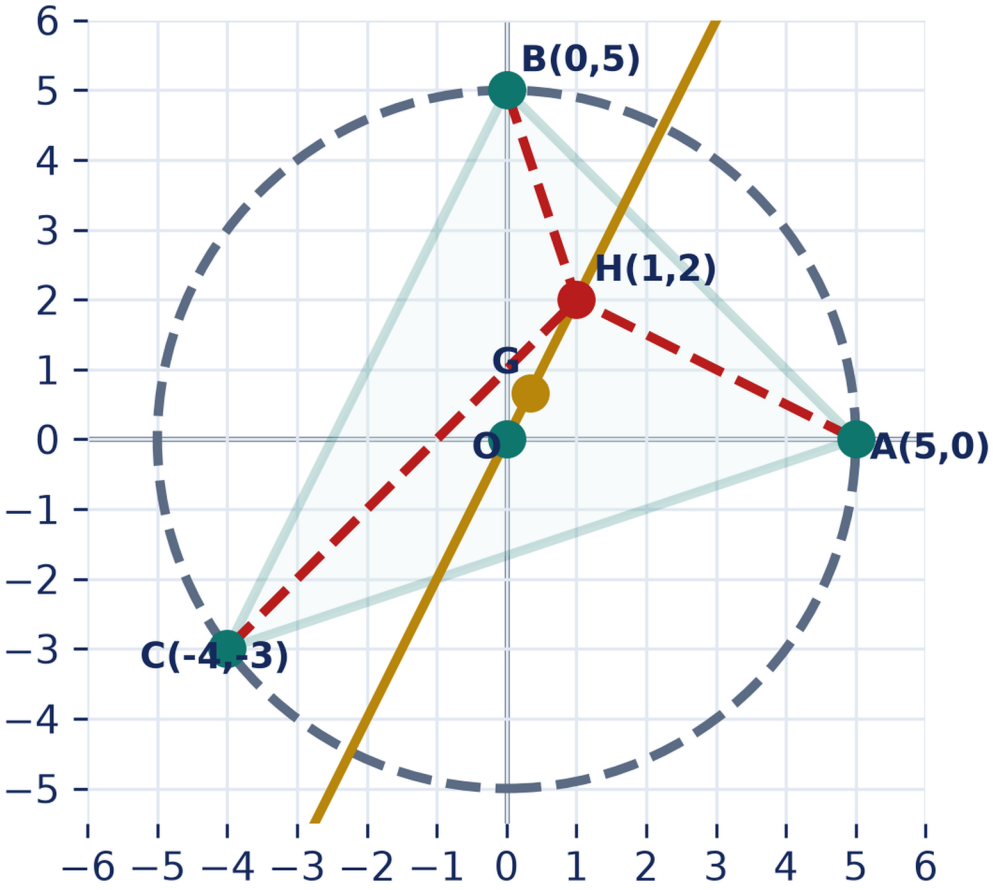
With a general circumcentre O, translate back to obtain $H = A + B + C - 2O$ and $G - O = (H - O)/3$.

COMMON TRAP

Equal distances from a sphere centre are not enough in 3D

For three points in space, a triangle's circumcentre must lie **in the triangle's plane** as well as be equidistant from its vertices. A sphere centre equidistant from all three need not lie in that plane. Before using $H = A + B + C$ with $O = 0$, verify both the equal lengths and that O belongs to the plane of A, B, C.

O, G and H lie on one Euler line



READ THE FIGURE
All three vertices lie on x^2+y^2=25. H=A+B+C=(1,2), G=H/3, and the dashed segments are altitudes.

Verify the example with lengths and dot products

For $A = (5, 0)$, $B = (0, 5)$ and $C = (-4, -3)$,

$$|A|^2 = |B|^2 = |C|^2 = 25,$$

so the origin is genuinely the circumcentre. Set $H = (1, 2)$. Then

$$(H - A) \cdot (C - B) = (-4, 2) \cdot (-4, -8) = 16 - 16 = 0,$$

$$(H - B) \cdot (A - C) = (1, -3) \cdot (9, 3) = 9 - 9 = 0.$$

The third altitude follows by the same calculation. Also $G = (1/3, 2/3) = H/3$, so

$$OG^2 = \tfrac{5}{9}, \qquad GH^2 = \tfrac{20}{9} = 4OG^2.$$

This verifies both collinearity and $OG : GH = 1 : 2$ without relying on the drawing.